



Ask and focus more: Question-prompt uncertainty allocation for dual-controllable video captioning

Shuqin Chen ^{a,b}, Xingrui Yang ^{c,1}, Yi Chen ^{d,e,1}, Kai Wang ^{a,f}, Xiaohan Yu ^g,
Xian Zhong ^{c,h,*}

^a School of Computer, Hubei University of Education, 430205, Wuhan, China

^b Hubei Provincial Collaborative Innovation Center for Basic Education Information Technology Services, Hubei University of Education, 430205, Wuhan, China

^c Hubei Key Laboratory of Transportation Internet of Things, School of Computer Science and Artificial Intelligence, Wuhan University of Technology, 430070, Wuhan, China

^d Key Laboratory of Advanced Medical Imaging and Intelligent Computing of Guizhou Province, Guizhou University, 550025, Guiyang, China

^e The D-Lab, Department of Precision Medicine, GROW-School for Oncology and Reproduction, Maastricht University, 6200 MD, Maastricht, Netherlands

^f School of Artificial Intelligence and Computer Science, Hubei Normal University, 435002, Huangshi, China

^g School of Computing, Macquarie University, 2109, Sydney, NSW, Australia

^h State Key Laboratory of Maritime Technology and Safety, Wuhan University of Technology, 430063, Wuhan, China

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ABSTRACT

Video captioning aims to generate natural language descriptions from video content via hierarchical architectures that capture key visual elements. Although entities, predicates, and syntactic structures are critical for coherent descriptions, existing controllable methods often struggle to dynamically emphasize salient information due to reliance on predefined entity lists and static fusion strategies. Moreover, uncertainty in aligning multimodal features (e.g., visual actions and textual semantics) further weakens model robustness. To address these challenges, we propose QPDC, a Dual-Controllable video captioning framework that introduces a Question-Prompt mechanism to adaptively guide caption focus. Unlike traditional entity-driven approaches, QPDC adopts a unified question-prompt strategy to maintain semantic consistency while enabling flexible attention shifts. Specifically, the Question-Prompt (Q-Prompt) module dynamically steers the model's focus according to query intent, emphasizing salient content without predefined constraints. The Selective Fusion Module (SFM) hierarchically integrates Q-Prompt guidance with local action features and global sentence context to support multi-granularity control. In addition, a Dirichlet-based strategy is employed to model uncertainty in multimodal weight allocation, improving adaptability under varying data conditions. Extensive experiments on MSVD and MSR-VTT show that QPDC achieves competitive performance among controllable video captioning methods and produces more diverse and user-preferred descriptions. Code will be available at <https://github.com/XiaotaiCHEN/QPDC.git>.

1. Introduction

Video captioning [1] aims to automatically generate natural language descriptions from videos and has been widely applied to assist visually impaired individuals, enhance human-computer interaction, and support video retrieval. However, without explicit contextual constraints, conventional video captioning models tend to produce relatively fixed descriptions driven primarily by dominant visual content. Despite recent progress in large language models, video captioning remains challenging, as videos often contain complex and diverse infor-

mation, of which only a small subset (e.g., two or three key elements) is relevant for concise and informative descriptions.

Controllable video captioning (CVC) extends this paradigm by incorporating explicit control conditions (e.g., questions or instructions) to generate customized and diverse descriptions that satisfy user-specified intents. In recent years, CVC has attracted increasing attention as a promising direction toward flexible and user-centered video understanding. A core challenge in CVC lies in detecting and integrating focused actions and semantics. Existing approaches can be broadly categorized into two classes. The first class relies on graph-based encoders to model

* Corresponding author.

E-mail addresses: csqcxw0801@163.com (S. Chen), yang_xingrui@whut.edu.cn (X. Yang), Yi_Chen.gzu@outlook.com (Y. Chen), wangkai_010722@163.com (K. Wang), xiaohan.yu@mq.edu.au (X. Yu), zhongx@whut.edu.cn (X. Zhong).

¹ Contributed equally to this work.

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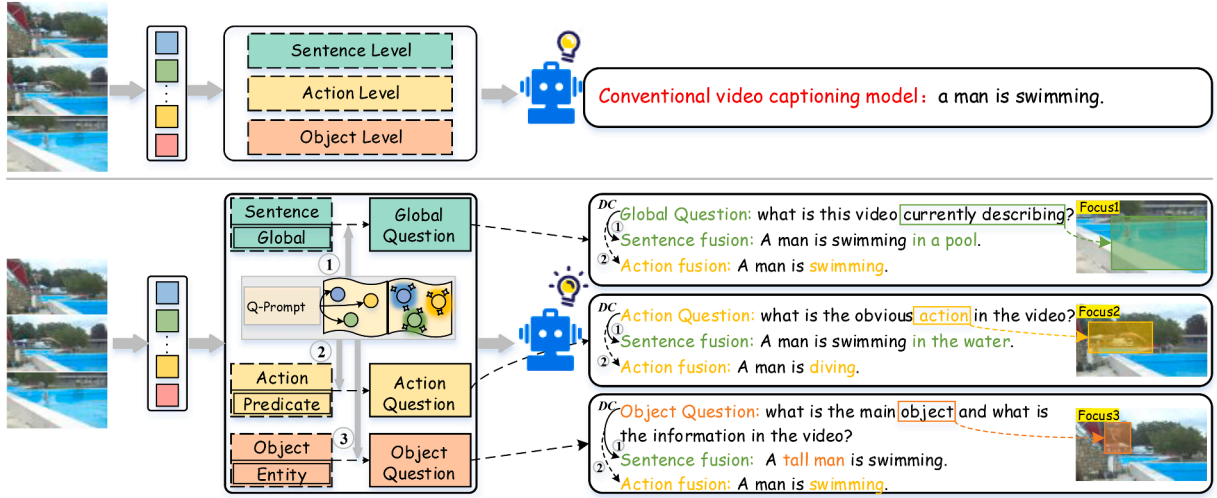


Fig. 1. Comparison of Traditional Methods and Our QPDC Approach. (a) Traditional controllable methods rely on sentence components and hierarchical structures, which often lead to rigid and inflexible control. (b) In contrast, QPDC employs question prompts to guide model focus across feature levels, enabling flexible and adaptive caption generation.

spatial and temporal relations among objects. For example, Chen et al. [2] design a spatiotemporal graph framework with recurrent region attention, cross-frame message passing, and a temporal graph decoder to capture high-order relations without external detectors. However, as observed in [3], such methods often generate generic descriptions and lack fine-grained controllability to emphasize specific points of interest or narrative patterns. To mitigate this issue, predicate-based feature selection and gated fusion mechanisms have been explored to support diverse generation under different control signals.

The second class focuses on reducing the semantic gap between video representations and linguistic captions prior to generation. GSEN [4], for instance, aligns global visual content with caption semantics, while IVRC [5] encodes object-, action-, and event-level features at multiple granularities under caption supervision. Although these methods investigate both local word alignment and global sentence alignment, they rarely address cross-granularity alignment between fine-grained actions and coarse-grained contextual semantics. Consequently, representative video embeddings may still overlook detailed actions or broader scene-level cues. We argue that question prompts provide an effective means to bridge this gap by explicitly guiding cross-granularity alignment. Notably, question-answer pairs lie at the core of visual question answering (VQA), which integrates visual understanding and language reasoning. While video captioning and VQA share common modeling components [6], we leverage questions not as queries to be answered, but as prompts that guide the model to focus on relevant visual information.

Prior studies have shown that question-driven supervision can benefit text generation. However, existing video captioning methods often lack sufficient textual guidance during visual feature learning, and the relation between questions and visual entities remains underexplored. For example, Hu et al. [7] introduce a question-prompt encoder to select relevant visual features from optical character recognition (OCR) regions and design targeted questions to ground caption generation. Nevertheless, the strong dependence on OCR features may induce shortcut learning. In contrast, our approach does not rely on pre-extracted OCR cues and instead focuses on generating fairness-aware questions to guide visual attention more generally. As illustrated in Fig. 1, captions produced by prior methods may fail to identify salient attributes (e.g., a “tall man”) or key actions (e.g., in a “pool”). From this perspective, video captioning can be viewed as answering an implicit question: “What does this video describe?” A high-quality caption should therefore respond to a range of related queries about the video, even in the absence of explicit visual input.

In this work, we propose **QPDC**, an encoder-decoder-based dual-controllable video captioning framework that leverages BLIP-driven questions, a Selective Fusion Module (SFM), and comprehensive focus modeling. QPDC generates captions by responding to video-relevant questions, thereby reducing omissions and semantic errors. Specifically, we generate visual questions from both videos and human-written captions and integrate them into a dedicated question-prompt (Q-Prompt) captioning model. To improve robustness under alignment uncertainty, we introduce branch-adjustable supervision that allows different branches (e.g., entity and action branches) to dynamically adjust their contributions based on input characteristics. The encoder jointly processes video content and questions, followed by an alignment module for feature fusion, while the decoder generates captions conditioned on the integrated representations. This design enables dynamic emphasis on specific actions or global semantics depending on the question context. Moreover, the fusion module supports sample-adaptive multimodal integration without additional computational overhead and performs evidential-level fusion via a learnable Dirichlet distribution.

The main contributions of this paper are summarized threefold:

- We propose QPDC, a dual-controllable video captioning framework that integrates question-compatible and branch-adjustable supervision through a unified sentence-level modeling paradigm.
- We design a Question Prompt (Q-Prompt) mechanism and a Selective Fusion Module (SFM) to enhance controllability. Q-Prompt generates BLIP-driven questions with global and local semantics, while SFM adaptively selects appropriate branches to support customized and diverse caption generation.
- We introduce a sample-adaptive multimodal integration strategy based on evidential modeling, enabling explicit uncertainty characterization and improving robustness under noisy or degraded conditions.

2. Related work

2.1. Video captioning

Early video captioning methods were largely template-based, detecting visual concepts such as objects, relations, and attributes and composing captions using predefined grammatical structures (e.g., subject-verb-object). With the emergence of encoder-decoder frameworks, MesNet [8] introduced a memory-sharing mechanism to strengthen

recurrent connections, while MAN [9] enhanced encoders with learnable visual memory vectors and incorporated external linguistic cues in the decoder. Subsequently, STAT [10] leveraged spatio-temporal attention to extract salient frames, and the T^2 Transformer [11] explored intra- and cross-modal semantic correlations to construct unified multi-modal representations.

The adoption of vision transformers [12] has further accelerated the transition toward transformer-based architectures. DAST [13] employed a depth-aware sparse transformer encoder to emphasize geometric relations, while VideoBLIP [14] adopted a transformer encoder-decoder pipeline built upon CNN features. However, many such approaches rely on frozen feature extractors or offline object detectors, which may introduce noise and limit end-to-end optimization. To address this issue, SwinBERT [15] proposed a fully transformer-based architecture with Video Swin Transformer encoders and a BERT-based [16] decoder.

With advances in multimodal learning [17], video captioning has increasingly emphasized cross-modal alignment. Li et al. [18] proposed a dual-branch MICM framework based on image-to-text and image-to-video alignment, while Fu et al. [19] introduced CLIP-AGIQA to mitigate domain gaps through multimodal prompt learning and vision-language consistency. Zero-shot video captioning has also attracted attention: Ma et al. [20] proposed a retrieval-enhanced test-time adaptation framework to improve robustness in open-domain settings. Other works incorporate textual supervision during visual encoding, jointly modeling object-, action-, and context-level features [6,21] or aligning video representations with linguistic semantics across multiple granularities [22]. As effective fusion relies heavily on alignment quality [19,23,24], CLIP-based methods [19] serve as representative solutions in this direction.

Despite these advances, question-prompt video captioning [7], which explicitly leverages questions to guide fine-grained and global semantic interactions, remains relatively underexplored. In contrast to alignment-only strategies, our work integrates question prompts with multi-granularity visual cues to support more controllable caption generation.

2.2. Question answering for visual captioning

Question answering has been shown to be beneficial for text generation. Shen et al. [25] reformulated captioning as a question-answering task to enhance semantic coverage. With the rapid development of large language models, recent studies [26] explore collaborative question answering between ChatGPT-like models and vision encoders for visually grounded caption generation. PromptCap [27], for example, employs GPT-3 to generate targeted questions for images and rewrites generic captions based on question-answer contexts using VQA data. Unlike these approaches, which rely on predefined or customized question-answer pairs, our method constructs fairness-aware questions and adopts a unified question-prompt structure to guide caption generation consistently across videos.

2.3. Controllable visual captioning

Our work is closely related to controllable captioning methods that incorporate explicit control signals [28]. Liu et al. [29] proposed O2NA, which identifies key objects and their attributes to generate drafts that are refined into fluent captions using contextual cues. Yang et al. [30] introduced masked scene graphs and masked autoregressive decoding to jointly model semantic and syntactic structures. Zhang et al. [31] incorporated part-of-speech tags as auxiliary inputs. Related efforts further explore controllability through subject-oriented encoding [32], POS-based visual switches [33], example-driven syntactic control [34], and two-stage style learning frameworks [35]. Other approaches leverage mouse traces, scene graphs, or length-level embeddings to guide caption generation.

Compared with these methods, we develop a fully attentive framework that supports dual controllability through question prompts and

adaptive fusion, while maintaining explicit grounding between visual content and textual semantics. Although related work [36] employs visual regions as control signals, it requires additional semantic annotations and introduces extra complexity, reflecting a different controllability setting from ours.

3. Proposed method

As shown in Fig. 2, given a video, we uniformly sample K keyframes and extract object regions and 2D appearance features P using pre-trained Faster R-CNN [37] and InceptionResNetV2 [38], respectively. Video objects are clustered and pooled by appearance similarity and intersection over union (IoU) to obtain initial object features D . In addition, adjacent frames around each keyframe are collected to form 3D cuboids, from which motion features M are extracted by a pre-trained C3D [39]. Following [22], we align video representations with language semantics at the object, predicate, and sentence levels.

3.1. Object module

The object module selects primary objects likely to appear in the caption. The initial object sequence D is first fed into a transformer encoder (without positional encoding) to obtain D' . The appearance features P and motion features M are concatenated and processed by a BiLSTM, followed by max-pooling to generate a video content embedding F . We then input T copies of F (i.e., $\{F\}_{\times T}$), Q randomly initialized query parameters, and D' into a transformer decoder to generate T primary object features:

$$O = \{o_i\}_{i=1}^T = \text{Dec}(D', Q, \{F\}_{\times T}). \quad (1)$$

For each noun in the ground-truth captions, we retrieve synonym labels using WordNet and encode the corresponding embeddings $E = \{e_i\}_{i=1}^L$ (excluding abstract nouns) with pre-trained SBERT [40]. A fully connected layer projects O into the language semantic space, yielding $O' = \{o'_i\}_{i=1}^T$. The embedding set E is padded with a special token ϕ (indicating no object) to match size T . The optimal assignment between E and O' is obtained by minimizing the distance cost:

$$\hat{e} = \arg \min_{\epsilon \in \Omega_T} \sum_{i=1}^T \text{Dist}(e_i, o'_{\epsilon(i)}), \quad (2)$$

where Ω_T is the assignment search space for T elements, and

$$\hat{e} = \arg \min_{\epsilon \in \Omega_T} \sum_{i=1}^T \text{Dist}(e_i, o'_{\epsilon(i)}), \quad (3)$$

where Ω_T is the assignment search space for T elements, and:

$$\text{Dist}(e_i, o'_{\epsilon(i)}) = \mathbb{1}_{\{e_i \neq \phi\}} \left(1 - \frac{e_i \cdot o'_{\epsilon(i)}}{\|e_i\| \cdot \|o'_{\epsilon(i)}\|} \right). \quad (4)$$

The object module is optimized by minimizing the total distance:

$$\mathcal{L}_o = \sum_{i=1}^T \text{Dist}(e_i, o'_{\epsilon(i)}). \quad (5)$$

3.2. Action module

The action module leverages primary object and motion features to construct action representations. A soft-attention mechanism first computes object-aware motion features $M^o = \{m_i^o\}_{i=1}^K$ from initial motion features $M = \{m_i\}_{i=1}^K$:

$$m_i^o = \text{SoftAttn}(m_i, o_j) = \sum_{j=1}^T \beta_{i,j} o_j, \quad (6)$$

where $\beta_{i,j} = \exp(\hat{\beta}_{i,j}) / \sum_{k=1}^T \exp(\hat{\beta}_{i,k})$ and:

$$\hat{\beta}_{i,j} = w_1^\top \tanh(W_1^\top m_i + W_2^\top o_j + b_1), \quad (7)$$

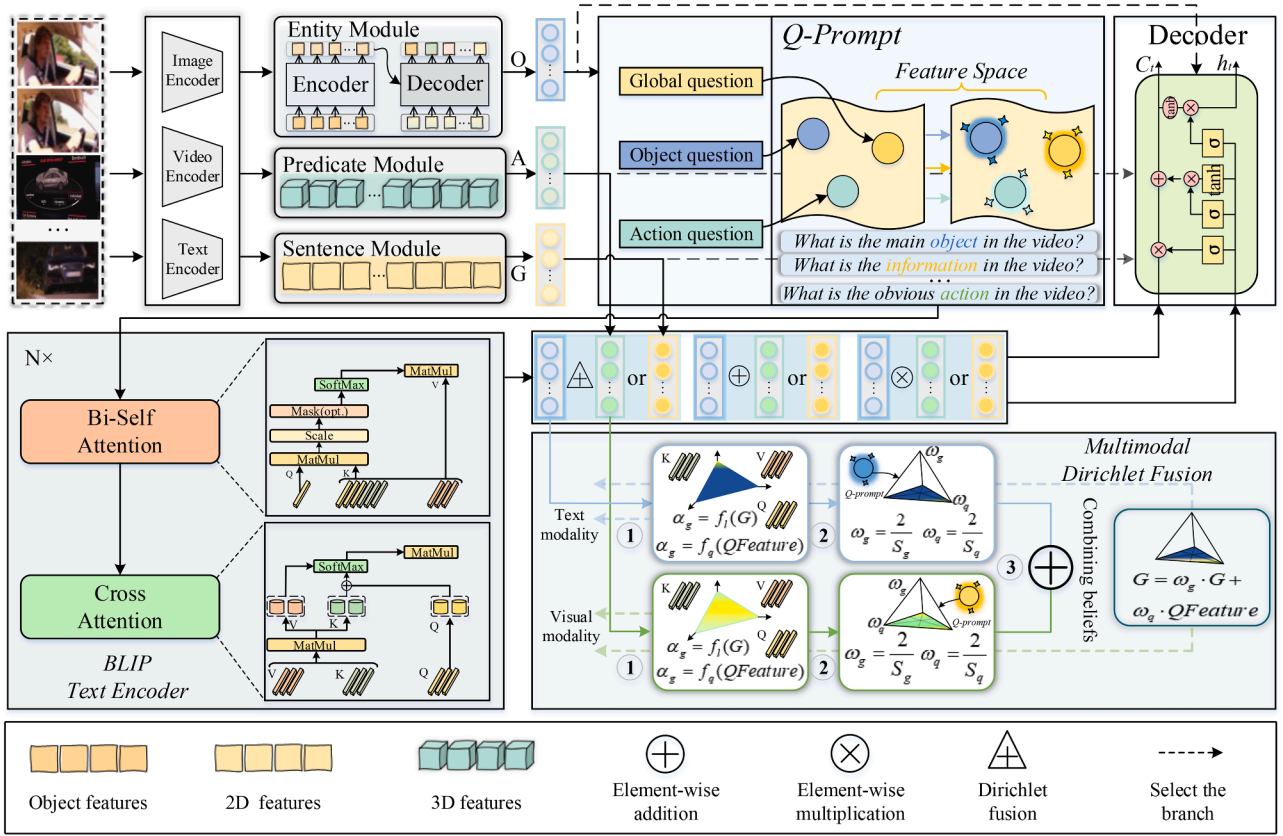


Fig. 2. Framework of the proposed QPDC. The framework comprises three modules: (1) the Question Prompt (Q-Prompt), which guides caption generation via different questions; (2) the Selective Fusion Module (SFM), which selectively merges question features with multi-granularity visual features to enhance diversity; and (3) Multiple Fusion Methods (MFM) for effective visual-text integration.

with w_1, W_1, W_2 , and b_1 learnable. A BiLSTM then encodes the concatenated pairs $\{[m_i; m_i^o]\}_{i=1}^K$ into action features:

$$A = \{a_i\}_{i=1}^K = \text{BiLSTM}(\{[m_i; m_i^o]\}_{i=1}^K). \quad (8)$$

A max-pooling operation aggregates A , and a fully connected layer projects the result into the language space as a' . Predicates in the ground-truth captions are encoded with SBERT as v . The action module is optimized via cosine distance:

$$\mathcal{L}_a = 1 - \frac{v \cdot a'}{\|v\| \cdot \|a'\|}. \quad (9)$$

3.3. Sentence module

The sentence module integrates object and action representations to construct a global video representation. Appearance-related object features $P^o = \{p_i^o\}_{i=1}^K$ and action-related features $P^a = \{p_i^a\}_{i=1}^K$ are computed via soft attention:

$$p_i^* = \text{SoftAttn}(p_i, P_j^*), \quad P^* \in \{P^o, P^a\}. \quad (10)$$

A BiLSTM then encodes $\{[p_i; p_i^a; p_i^o]\}_{i=1}^K$ into global features:

$$G = \{g_i\}_{i=1}^K = \text{BiLSTM}(\{[p_i; p_i^a; p_i^o]\}_{i=1}^K). \quad (11)$$

After max-pooling G , a fully connected layer projects it into the language space as g' . The full ground-truth caption is encoded using SBERT as c , and the sentence module is optimized by:

$$\mathcal{L}_g = 1 - \frac{c \cdot g'}{\|c\| \cdot \|g'\|}. \quad (12)$$

3.4. Question-prompt encoder module

To guide caption generation before decoding, we introduce a question-prompt mechanism. The CLIP text encoder, BLIP text encoder, and multimodal large model mPLUG-2 [41] are used to encode questions, as shown in Fig. 3(b,c). The BLIP tokenizer converts a natural language question into tokens:

$$QToken = \text{BLIPTokenizer}(\text{Question}). \quad (13)$$

The question (e.g., “What is this video describing?”, “What is the main object and what information is conveyed in the video?”, “What is the obvious action in the video?”) is processed with the BLIP text encoder and object features O to produce question representations:

$$QFeature = \text{BLIPTextEncoder}(QToken, O). \quad (14)$$

The text encoder applies Bi-Self-Attention, Cross-Attention, and feed-forward layers:

$$Q_s = \text{Bi-SelfAttn}(QToken), \quad (15)$$

$$Q_c = \text{CrossAttn}(Q_s', O), \quad (16)$$

$$Q_f = \text{FFN}(Q_c'), \quad (17)$$

$$QFeature = Q + Q_s + Q_c + Q_f. \quad (18)$$

The resulting question features guide action and global representations via element-wise multiplication:

$$A = A \cdot QFeature, \quad G = G \cdot QFeature. \quad (19)$$

3.5. Multiple fusion methods

Beyond multiplicative fusion, we introduce additive and Dirichlet-based fusion to provide fine-grained semantic guidance, as shown in

Table 1

Comparison of results with state-of-the-art methods on MSR-VTT and MSVD. Bold numbers indicate the best performance.

Method	Venue	MSR-VTT				MSVD			
		B-4	M	R	C	B-4	M	R	C
Pre-training Video Captioning									
HRNAT [6]	TIP'21	42.1	28.0	61.6	48.2	55.7	36.8	74.1	98.1
MGCMP [2]	ICCV'21	41.7	28.9	62.1	51.4	55.8	36.9	74.5	98.5
OpenBook [44]	CVPR'21	42.8	29.3	61.7	52.9	–	–	–	–
SHAN [45]	TCSVT'22	39.7	28.3	60.4	49.0	54.3	35.3	72.7	91.3
LSRT [46]	TIP'22	42.6	28.3	61.0	49.5	55.6	37.1	73.5	98.5
R-ConvED [47]	TOMM'23	40.4	28.1	–	47.9	53.5	34.6	–	82.4
RSFD [48]	AAAI'23	43.4	29.3	62.3	53.1	51.2	35.7	72.9	96.7
CoCap [49]	ICCV'23	43.1	29.8	62.7	56.2	55.9	39.9	76.8	113.0
MAN [9]	TMM'24	41.3	28.0	61.4	49.8	59.7	37.3	74.3	101.5
GSEN [4]	PR'24	42.9	28.4	61.7	51.0	58.8	37.6	75.2	102.5
ALSO-Net [21]	TOMM'24	41.9	28.9	62.0	51.3	55.7	35.9	73.0	89.0
RLHMN [22]	TPAMI'24	45.1	28.8	63.6	54.2	59.9	36.2	74.2	104.7
VCRN [50]	TNNLS'25	41.5	28.1	61.2	50.9	59.1	37.4	74.6	100.8
Controllable Video Captioning									
O2NA [29]	ACL'21	41.6	28.5	62.4	51.1	–	–	–	–
MSG [30]	PRICAI'22	41.6	28.9	62.8	51.2	55.6	37.7	74.6	96.6
Baseline		41.78	28.61	61.49	49.87	58.35	36.56	73.53	99.85
QPDC (Ours)		42.42	28.93	61.86	51.79	59.36	38.35	74.84	105.19

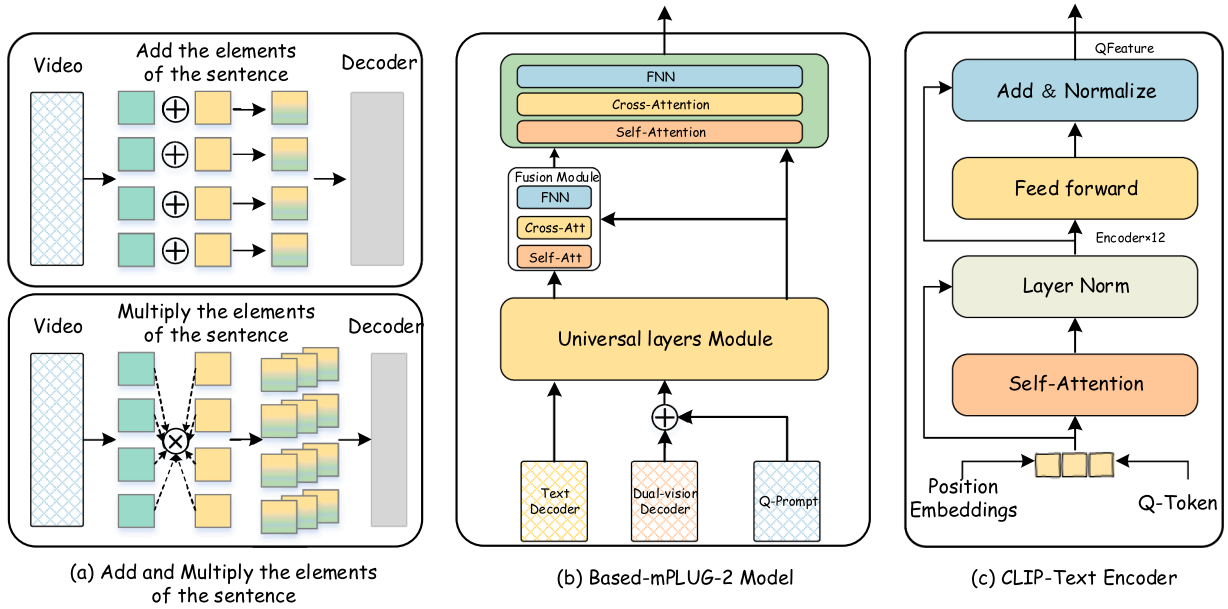


Fig. 3. Variants for local semantic guidance to the caption encoder. (a) Additive fusion of question and visual features; (b) multiplicative fusion; (c) Q-Prompt encoding based on the CLIP text encoder.

Table 2

Performance comparison on VATEX.

Method	Venue	B-4	M	R	C
OpenBook [44]	CVPR'21	33.9	23.7	50.2	57.5
CLIP-Base [51] †	PRCV'22	36.8	24.8	51.9	60.9
HMN [52]	CVPR'23	33.2	22.7	49.3	51.6
R-ConvED [47]	TOMM'23	32.1	21.8	–	48.7
IVRC [5]	PR'24	32.8	24.0	50.3	57.4
ALSO-Net [21]	TOMM'24	27.8	20.6	46.8	40.4
QPDC (Ours)		36.8	24.9	52.0	61.3

Fig. 3(a):

$$A = A + \text{QFeature}, \quad G = G + \text{QFeature}. \quad (20)$$

A Dirichlet distribution is further used to model multimodal uncertainty. Instead of sampling, we use the Dirichlet expectation to compute

fusion weights for stable and normalized allocation during inference:

$$G = w_g \cdot G + w_q \cdot \text{QFeature}, \quad (21)$$

$$w_g = \frac{\alpha_g}{\alpha_g + \alpha_q}, \quad w_q = \frac{\alpha_q}{\alpha_g + \alpha_q}, \quad (22)$$

$$\alpha_g = \text{Softplus}(f_l(G)) + \epsilon, \quad (23)$$

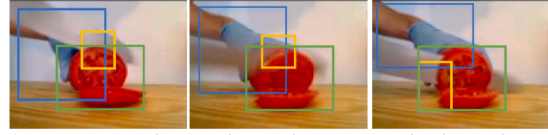
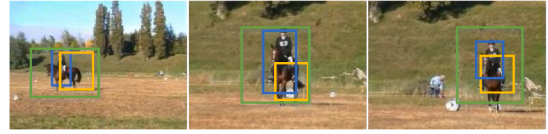
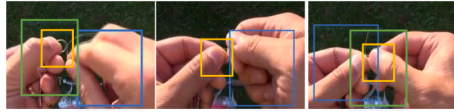
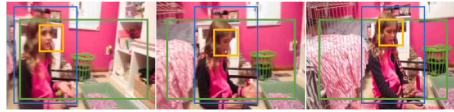
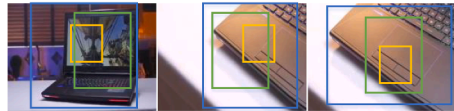
$$\alpha_q = \text{Softplus}(f_l(\text{QFeature})) + \epsilon, \quad (24)$$

where w_g and w_q are fusion weights, α_g and α_q are Dirichlet parameters, and f_l is a linear projection. Softplus ensures $\alpha > 0$, and $\epsilon \approx 10^{-6}$ avoids numerical instability. We validate that the Dirichlet-expectation weights align model confidence with empirical accuracy, and the confidence correlates positively with caption quality (Pearson correlation > 0.85).

Table 3

Ablation study on MSR-VTT and MSVD. Bold values indicate the best performance.

Baseline	Q-Prompt	SFM Action	SFM Sentence	MSR-VTT				MSVD			
				B-4	M	R	C	B-4	M	R	C
•	◦	◦	◦	41.78	28.61	61.49	49.87	58.35	36.56	73.53	99.85
•	•	◦	•	42.42	28.93	61.86	51.79	58.02	36.85	74.28	104.27
•	•	•	◦	43.41	28.26	61.80	50.53	59.36	38.35	74.84	105.19

**Question:** What is this video currently describing?**GT:** a baby is playing with a dog.**Baseline:** a dog is playing.**Ours:** a baby is playing with a dog.**Question:** What is this video currently describing?**GT:** a woman is slicing tomato.**Baseline:** a person is slicing tomato.**Ours:** a woman is slicing tomato.**Question:** What is this video currently describing?**GT:** a man jumping rope.**Baseline:** a boy is running.**Ours:** a man is jumping.**Question:** What is this video currently describing?**GT:** a woman is riding a horse.**Baseline:** a man is riding a horse.**Ours:** a woman is riding a horse.**Fig. 4.** First choice of SFM: fusion of question features and action features A. Comparison between the baseline and QPDC on MSVD.**Question:** What is this video currently describing?**GT:** a person is making a coil with a thread.**Baseline:** a man is holding a piece of plastic.**Ours:** a person is showing how to use a toy.**Question:** What is this video currently describing?**GT:** a cartoon is talking in space.**Baseline:** a cartoon character is flying.**Ours:** a cartoon character is talking.**Question:** What is this video currently describing?**GT:** small girl is sitting and talking in her room.**Baseline:** a woman is talking.**Ours:** a girl is sitting in a room.**Question:** What is this video currently describing?**GT:** a woman is walking around town.**Baseline:** there is a woman is walking on the street.**Ours:** there is a woman is walking.**Question:** What is this video currently describing?**GT:** a man is talking and a laptop is there.**Baseline:** a computer screen is being shown.**Ours:** a man is talking about a laptop.**Question:** What is this video currently describing?**GT:** a person is playing a video game.**Baseline:** a man is talking about a computer program.**Ours:** a person is playing a video game.**Fig. 5.** Second choice of SFM: fusion of question features and sentence features G. Comparison between the baseline and QPDC on MSR-VTT.

3.6. Caption decoder

The caption decoder sequentially generates captions by integrating multi-level video representations, their language projections, and the previous word embedding w_{t-1} :

$$h_t = \text{LSTM}([\bar{g}_t; \bar{a}_t; \bar{o}_t; w_{t-1}], h_{t-1}), \quad (25)$$

where

$$\bar{g}_t = [g_t^h; g_t'], \quad \bar{a}_t = [a_t^h; a_t'], \quad \bar{o}_t = [o_t^h; o_t'], \quad (26)$$

and g_t^h , a_t^h , o_t^h , and $o_t'^h$ are computed via soft attention using the previous hidden state h_{t-1} and corresponding features $\{g_k\}$, $\{a_k\}$, $\{o_k\}$, and $\{o_k'\}$:

$$C_t^h = \text{SoftAtten}(h_{t-1}, C_k), \quad (27)$$

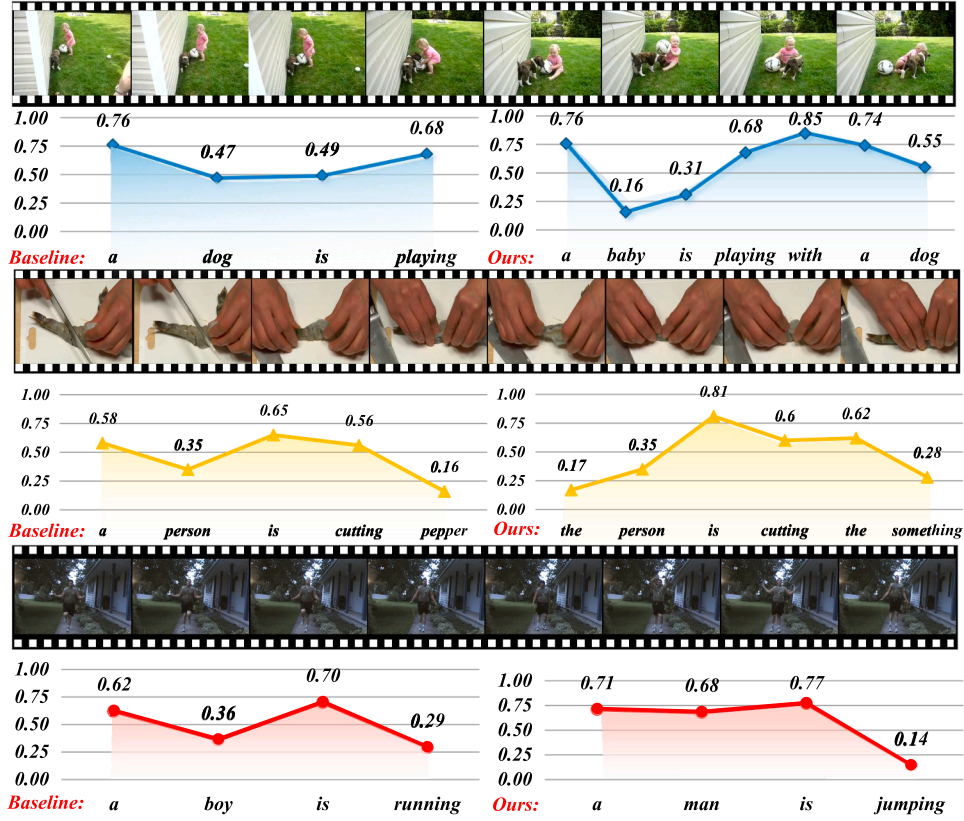


Fig. 6. Comparison of word confidence in captions generated by the baseline and QPDC.

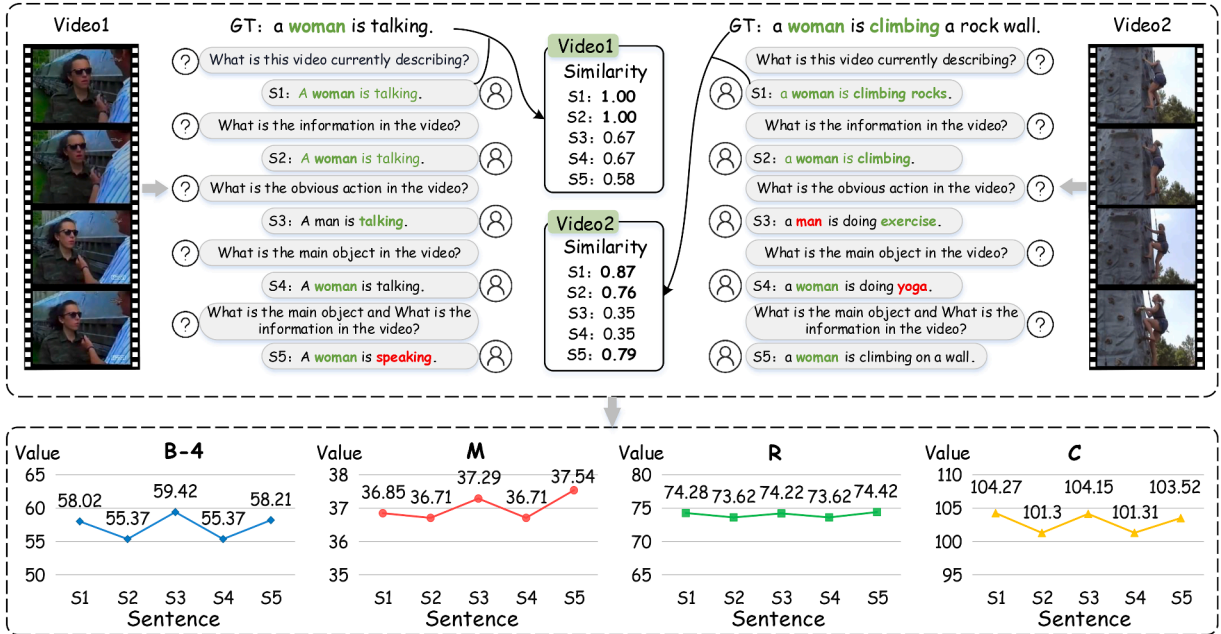


Fig. 7. First choice of SFM: caption evaluation and performance comparison. Top: qualitative comparison under different question prompts and ground truth. Bottom: quantitative evaluation for different question-prompt variants.

with $C \in \{g_k, a_k, o_k, o'_k\}$. The hidden state h_t is then fed into a fully connected layer and a softmax function to obtain the vocabulary distribution $\mathcal{P}_t$.

Given the ground-truth caption $[x_1, x_2, \dots, x_{L_c}]$ of length L_c , the cross-entropy loss is:

$$\mathcal{L}_{\text{lang}} = - \sum_{t=1}^{L_c} y_t^\top \log \mathcal{P}_t, \quad (28)$$

where y_t is the one-hot word vector. We further optimize $\mathcal{L}_{\text{soft}}$ from an external language model (ELM) to transfer linguistic knowledge to the decoder and alleviate long-tail word distribution issues [22].

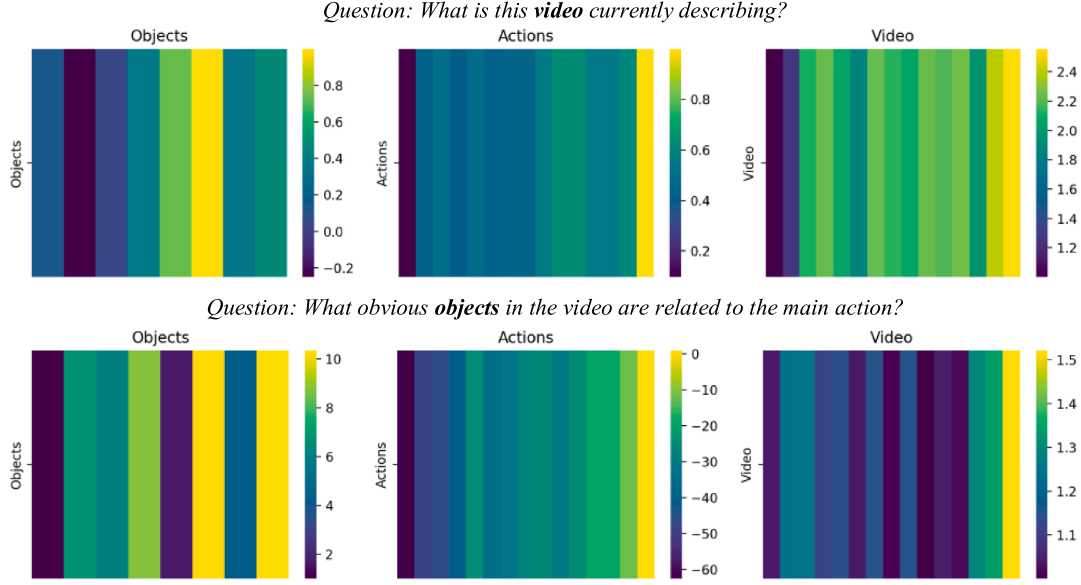


Fig. 8. Attention shift heatmap.

Table 4

Comparison of fusion methods between question and visual features. + denotes additive fusion, * denotes multiplicative fusion, and DS denotes Dirichlet-based fusion.

Method	MSR-VTT				MSVD			
	B-4	M	R	C	B-4	M	R	C
Baseline	41.78	28.61	61.49	49.87	58.35	36.56	73.53	99.85
Question Sentence DS	42.39	28.63	61.86	51.45	57.67	37.18	73.78	102.83
Question Sentence +	42.52	28.91	62.99	51.37	56.86	36.90	73.88	103.55
Question Sentence *	42.42	28.93	61.86	51.79	58.02	36.85	74.28	104.27

Table 5

Text encoder replacement on MSR-VTT and MSVD.

Text Encoder	MSR-VTT				MSVD			
	B-4	M	R	C	B-4	M	R	C
CLIP Text Encoder	40.99	27.44	60.32	48.46	57.40	37.15	73.40	102.21
BLIP Text Encoder	42.42	28.93	61.86	51.79	59.36	38.35	74.84	105.19

The final loss is:

$$\mathcal{L} = \lambda_o \mathcal{L}_o + \lambda_a \mathcal{L}_a + \lambda_g \mathcal{L}_g + \lambda_{\text{lang}} \mathcal{L}_{\text{lang}} + \lambda_{\text{soft}} \mathcal{L}_{\text{soft}}, \quad (29)$$

where λ_o , λ_a , and λ_g are balancing coefficients for the entity-, action-, and sentence-level objectives, respectively. To avoid excessive hyperparameter tuning and improve training stability, λ_{lang} and λ_{soft} are fixed to 1 in all experiments, and we focus on optimizing the first three coefficients.

4. Experimental results

4.1. Experimental settings

We evaluate our method on two widely used video captioning benchmarks. MSR-VTT [42] contains 6513 training, 497 validation, and 2990 test clips (15s each), with 20 human-annotated captions per clip across 20 categories. MSVD [43] includes 1970 YouTube videos with up to 40 English captions each, split into 1,200/100/670 for training/validation/testing. Following standard practice, we report BLEU-4 (B-4), METEOR (M), CIDEr (C), and ROUGE (R).

For each video, we sample 15 clips of 16 consecutive frames. Object regions are detected in the 8th frame using Faster R-CNN [37], while 2D appearance features are extracted by InceptionResNetV2 [38]. Motion features are obtained using a pre-trained C3D [39] on Kinetics-400. Questions are encoded via BLIP with padding length 15 and hidden size 512.

We fine-tune the model following [22]. The loss weights λ_o , λ_a , λ_g , λ_{lang} , and λ_{soft} are set to 0.6, 0.3, 1.0, 0.5, and 0.5, respectively. We adopt Adam with a learning rate of 1×10^{-4} , batch size 64, and train for 20 epochs. The decoder hidden size is 512. Beam search with beam size 5 is applied during inference. All experiments are conducted with PyTorch on an RTX-3090Ti (24 GB).

4.2. Comparison with state-of-the-art methods

On MSR-VTT, our method achieves competitive results across all metrics (BLEU-4, METEOR, ROUGE, and CIDEr), with clear gains on BLEU-4 and CIDEr. On MSVD, the improvement is more pronounced, reaching a CIDEr score of 105.19. These results support the effectiveness of question-driven guidance for controllable captioning. In particular, QPDC consistently improves over prior controllable video captioning methods under the same evaluation protocol.

Table 1 compares QPDC with representative models on both datasets. QPDC yields consistent gains over controllable baselines, especially on CIDEr: on MSR-VTT, it improves by +1.4% over O2NA [29] and +1.2% over MSG [30]. Compared with RLHMN [22], QPDC is higher on MSVD but slightly lower on MSR-VTT. This difference is expected given their distinct objectives: RLHMN targets general video captioning with end-to-end cross-modal semantic alignment, whereas QPDC is tailored to dual-controllable captioning, emphasizing question-prompt guidance and uncertainty-aware fusion for multi-granularity control. Under this controllable setting, QPDC demonstrates clear advantages in intent alignment, diversity, and controllability, and achieves competitive performance on MSVD and MSR-VTT.

To evaluate the effectiveness of the proposed Q-Prompt model, we conduct an ablation study on VATEX. As shown in Table 2, integrating Q-Prompt consistently improves performance, demonstrating its broad applicability. VATEX contains longer and more complex scenes, and its evaluation metrics emphasize fluency and comprehensiveness more than question-specific controllability.

Table 6
Second choice of SFM: impact of different question prompts on MSR-VTT and MSVD.

Variant Questions	MSR-VTT				MSVD			
	B-4	M	R	C	B-4	M	R	C
What is the obvious action in the video?	42.35	27.55	60.71	49.34	57.72	37.23	74.25	103.06
What is the main object in the video?	42.35	28.02	61.56	49.69	58.75	37.57	74.63	103.35
What is the information in the video?	43.07	27.98	61.67	49.91	58.21	37.45	74.78	104.99
What is this video currently describing?	42.42	28.93	61.86	51.79	59.36	38.35	74.84	105.19

Table 7
Quantitative comparison of question-prompt strategies on MSVD.

Methods	B-4	M	R	C
Baseline	58.3	36.6	73.5	99.9
What is this video currently describing?	59.4	38.4	74.8	105.2
What obvious objects in the video are related to the main action?	61.7	38.9	75.2	109.7

Table 8
Diversity evaluation on MSVD.

Method	Div-1 $\uparrow$	Div-2 $\uparrow$
Baseline	93.84	99.67
QPDC (Ours)	94.08	99.84

4.3. Ablation studies

4.3.1. Selective fusion module

We ablate the dual controllable branches (action-level and sentence-level) on MSVD and MSR-VTT. As shown in Table 3, both branches contribute to performance gains. On MSR-VTT, the sentence-level branch improves BLEU-4, METEOR, ROUGE, and CIDEr by 1.5%, 1.1%, 0.6%, and 3.9%, respectively. On MSVD, the action-level branch yields gains of +1.7%, +4.9%, +1.8%, and +5.3%. These results support the effectiveness of SFM and the Q-Prompt encoder in enhancing compositional grounding and semantic precision.

Fig. 4 visualizes predictions under ablations. While ablated variants remain reasonable, they often miss key semantics. For example, the baseline outputs “a dog is playing”, whereas QPDC produces “a baby is playing with a dog”, reflecting improved focus on core elements and their relations. Fig. 5 further provides qualitative comparisons with visual grounding: our model distinguishes “girl” from “woman” and avoids confusing background objects (e.g., “computer”).

4.3.2. Fusion methods and representation choices

Table 4 compares different fusion strategies. Our MFM (Dirichlet, additive, and multiplicative fusion) consistently improves feature integration. Both additive and multiplicative fusion [7] outperform direct fusion [5].

Table 5 compares BLIP and CLIP text encoders. Replacing BLIP with CLIP (ViT-B/32) yields varied results, while BLIP generally performs better due to deeper cross-attention-based vision-language fusion.

4.3.3. Confidence score analysis

Fig. 6 compares caption confidence distributions between the baseline and QPDC. While the baseline predicts “a dog is playing”, QPDC outputs “a boy is playing with a dog” with higher confidence and correct age-related reasoning. In ambiguous scenes, QPDC integrates complementary cues to produce more grounded and coherent descriptions.

4.4. Guiding question analysis

4.4.1. Sentence-based captioning model

We analyze question specificity by varying the guiding prompt. As shown in Fig. 7, general prompts (e.g., “What is this video currently

describing?”) encourage broader alignment with ground-truth semantics. In contrast, specific prompts (e.g., “What is the obvious action in the video?”) sharpen action awareness, producing focus-driven captions such as “A man is doing exercise” by highlighting climbing motion cues. This indicates that prompt formulation directly affects semantic emphasis.

4.4.2. Action-based captioning model

As reported in Table 6, SFM enables Q-Prompt to inject action-aware signals into caption generation, and all question-guided variants outperform the baseline. Using the global prompt “What is this video currently describing?”, QPDC improves CIDEr by +3.9% on MSR-VTT and +5.3% on MSVD, supporting improved action-language coherence.

4.4.3. Qualitative experiments on attention shift

In Table 7, we evaluate controllable variants with different prompts to assess how question specificity affects performance. As shown in Fig. 7, when questions contain explicit elements (e.g., actions or objects), generated captions more strongly reflect those elements, indicating effective attention steering.

To further analyze attention shifts, we provide heatmap visualizations in Fig. 8. When a question emphasizes a specific object, the fused features exhibit higher similarity to object-level features; when it targets the video as a whole, the fused features align more with global features. This confirms that varying prompts can trigger attention shifts and support dual-controllable caption generation.

4.4.4. Diversified indicator evaluation

To assess controllability from a diversity perspective, we report Div-1 and Div-2 in Table 8, measuring lexical richness and n-gram diversity. QPDC achieves higher diversity scores than the baseline, indicating stronger generative flexibility. Combined with the prompt-controlled analyses in Fig. 7, these results suggest improved controllability through question-guided emphasis shifts.

4.4.5. Generalization ability on multimodal large language models

Inspired by prompt-based learning in large models [53], we further evaluate compatibility with mPLUG-2 [41] on MSR-VTT using the prompt “What is this video currently describing?”. As shown in Table 9, Q-Prompt improves most metrics on MSR-VTT and yields notable CIDEr gains on MSVD using a single A800 GPU, demonstrating generalization and adaptability.

4.4.6. FLOPs / Latency / Memory comparison

We report FLOPs, inference latency, and GPU memory usage in Table 10. Compared with the baseline model, QPDC introduces additional computational overhead. This overhead mainly stems from the BLIP-based question encoder, which involves multiple cross-attention operations, as well as the selective fusion module that incorporates additional

Table 9
Ablation of the Q-prompt encoder on the multimodal large language model mPLUG-2.

Method	Venue	MSR-VTT				MSVD			
		B-4	M	R	C	B-4	M	R	C
mPLUG-2 [41]	ICML'23	23.40	25.38	47.71	30.89	32.46	33.46	59.87	78.19
mPLUG-2 + Q-Prompt		24.50	24.68	48.71	36.40	31.76	32.80	59.68	81.80

Table 10
Quantitative analysis on MSVD. † indicates reproduced results.

Method	Venue	FLOPs	Latency	Memory
HMN [52] w/o Controller †	CVPR'23	163.2 G	36.2 ms	3.58 G
QPDC w/ Controller (Ours)		309.9 G	2495.8 ms	4.15 G

Table 11
Quantitative comparison of Dirichlet-based fusion variants on MSVD.

Method	B-4	M	R	C
Baseline	58.35	36.56	73.53	99.85
QPDC + Xavier Initialization	57.68	37.53	74.35	104.12
QPDC (Default)	59.36	38.35	74.84	105.19

Table 12
Sensitivity analysis of loss weights on MSVD.

λ_o	λ_a	λ_g	B-4	M	R	C
0.5	0.3	1.0	57.83	37.38	73.92	103.68
0.6	0.3	1.0	59.36	38.35	74.84	105.19
0.7	0.3	1.0	56.98	37.00	73.54	100.03
0.6	0.2	1.0	56.39	36.42	73.74	101.21
0.6	0.3	1.0	59.36	38.35	74.84	105.19
0.6	0.4	1.0	57.51	37.38	73.93	105.83
0.6	0.3	0.9	56.30	36.86	73.92	100.27
0.6	0.3	1.0	59.36	38.35	74.84	105.19

learnable linear projections. These components increase both computation and memory consumption.

Despite the increased cost, the proposed design delivers clear gains in controllability, semantic alignment, and caption diversity, representing a deliberate trade-off between performance and efficiency. Importantly, the backbone architecture (Transformer- and LSTM-based) is kept unchanged, and the additional overhead arises from the controller modules rather than the backbone itself. This analysis motivates future work on optimizing the controller design and reducing spatio-temporal redundancy to improve efficiency while preserving controllability.

4.4.7. Ablation and hyperparameter analysis

We further conducted extended ablation studies by varying the initialization of the evidence prediction layer in the Dirichlet-based fusion mechanism. In particular, we examined a Xavier-uniform Dirichlet initialization for comparison. As shown in Table 11, the default initialization adopted in our model consistently outperforms the Xavier-based Dirichlet initialization, thereby demonstrating the effectiveness and robustness of the proposed Dirichlet fusion design.

In addition, the total loss function contains multiple weighting hyperparameters that balance different learning objectives. To assess model stability, we conduct a sensitivity analysis by varying one hyperparameter at a time while fixing the others. As shown in Table 12, adjusting λ_o and λ_a within a reasonable range leads to smooth performance variations, with the default setting ($\lambda_o = 0.6$, $\lambda_a = 0.3$, $\lambda_g = 1.0$) consistently achieving strong or optimal results across all metrics. We further observe that moderate changes around this configuration do not cause abrupt performance degradation, indicating stable optimization behavior. Similarly, slight perturbations of λ_g produce comparable trends. These results suggest that the proposed framework is not overly sensi-

tive to precise hyperparameter tuning and exhibits robust performance across a broad range of settings.

5. Conclusion

In this paper, we propose QPDC, a dual-controllable video captioning framework that integrates a Question Prompt (Q-Prompt) and a Selective Fusion Module (SFM) to generate precise and semantically focused captions. Q-Prompt leverages BLIP-based prompt encoding to guide attention toward salient content. Experiments on MSVD and MSR-VTT demonstrate the effectiveness of QPDC, and qualitative results further confirm improved semantic grounding under controllable inputs.

We note that the current design still depends on predefined prompts and does not yet explore fully dynamic question patterns, which may limit generalization in open-ended scenarios. In addition, the Dirichlet-based fusion strategy warrants deeper analysis under extreme uncertainty conditions. Future work will investigate dynamic prompt optimization, e.g., via meta-learning or reinforcement learning, to enable more adaptive and context-aware controllability.

CRedit authorship contribution statement

Shuqin Chen: Writing – review & editing, Writing – original draft; **Xingrui Yang:** Validation; **Yi Chen:** Resources, Methodology; **Kai Wang:** Visualization; **Xiaohan Yu:** Supervision, Formal analysis; **Xian Zhong:** Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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